

Methods: ngc_3256

ALMA Thesis Planner (automated pipeline)

1. Data acquisition and organization (week 1). Query the ALMA Science Archive (<https://almascience.nrao.edu>) for project code 2017.1.00379.S and download the QA2 image products for the three member OUSs that define the science scope: ‘uid://A001/X1273/Xb55’ (Band 5, $\sim 7.0''$), ‘uid://A001/X1273/Xb4f’ (Band 6, $\sim 4.4''$), and ‘uid://A001/X1273/Xb47’ (Band 7, $\sim 3.2''$). Also download the QA2 products for ‘uid://A001/X1273/Xb4d’ (Band 6, $\sim 0.8''$) and ‘uid://A001/X1273/Xb4b’ (Band 6, $\sim 0.24''$), which are used only in steps 6 and 7. Do not download the raw measurement sets unless step 3 forces a re-imaging exception. Set up a version-controlled working directory with a fixed layout (one subdirectory per member OUS, one per analysis step) and a lab-notebook file in which every cube, script, and derived number is logged with the date and the script that produced it. All analysis is scripted in Python using ‘astropy’, ‘spectral-cube’, and ‘numpy’/‘matplotlib’; CASA (any recent version, e.g. 6.x) is used for image-plane operations (‘imsmooth’, ‘imsubimage’, ‘imstat’, ‘specflux’, ‘imcontsub’). Nothing is done interactively that cannot be reproduced by rerunning a script.

Before touching the cubes, build the line bookkeeping table. Verify every rest frequency against Splatalogue (HCN 2–1 at 177.261 GHz, 3–2 at 265.886 GHz, 4–3 at 354.505 GHz; HCO+ 2–1 at 178.375 GHz, 3–2 at 267.558 GHz, 4–3 at 356.734 GHz; H¹³CN 4–3 at 345.340 GHz; CS 7–6 at 342.883 GHz). Adopt $v_{\text{sys}} = 2775$ km/s (radio convention, LSRK) as the working systemic velocity — record the exact value and convention you adopt, and use it identically everywhere. For each line, compute the observed frequency and the channel range corresponding to $v_{\text{sys}} \pm 600$ km/s in each cube, and tabulate which spectral window contains it and how far each edge of that velocity range sits from the window edge.

2. Go/no-go gate (weeks 1–2). Three checks, executed in order, each producing a written pass/fail entry in the notebook.

Check A — Band 5 coverage. HCN(2–1) and HCO+(2–1) fall near window edges (~ 175.6 and ~ 176.7 GHz observed). Using the cube headers (‘CRVAL3’, ‘CDELTA3’, ‘NAXIS3’) and the bookkeeping table, confirm that valid (non-blanked, non-edge-flagged) channels cover at least $v_{\text{sys}} \pm 400$ km/s for both lines. Inspect the per-channel rms as a function of frequency (‘imstat’ per plane, or ‘spectral-cube’ ‘mad_std’ per channel) and flag any edge channels where the noise rises by more than $\sim 30\%$ above the median; those channels are excluded from the usable range. Coverage passes only if the ± 400 km/s range survives this exclusion for both J=2–1 lines.

Check B — continuum-subtraction quality. NGC 3256’s nuclei are bright in continuum, and a baseline residual can mimic or hide faint wings. For each of the three science cubes, extract a spectrum through the aperture defined below and inspect the line-free channels ($|v - v_{\text{sys}}| > 600$ km/s, away from all tabulated lines). Fit a zeroth- or first-order polynomial to those channels; if the fitted offset or slope is significant compared to the channel noise ($> 1\sigma$ per channel systematically), subtract the fit in the image plane (per-pixel, using ‘imcontsub’ with order 0 or 1 over the same line-free channel set) and re-verify. Higher-order fits are not permitted — if order 1 does not flatten

the baseline, that cube is escalated to me as potentially unusable, which is the only condition under which re-imaging from the archived measurement sets is considered, and any such exception gets its own documented appendix.

Check C — J=2-1 wing detection. Convolve all three science cubes to a common circular beam slightly larger than the Band 5 beam (adopt 7.5" round; use ‘imsmooth’ with ‘targetres=True’) and regrid them onto a common velocity grid (10 km/s channels, LSRK, radio convention) with ‘spectral-cube’. Define once, and never change afterward: a single circular aperture of radius 7.5" centered at the midpoint between the two nuclei (centered on the phase center at 156.96346°, -43.90461° unless the continuum peaks say otherwise; record the final center). Extract the aperture-integrated spectrum of HCN(2-1) and HCO+(2-1). Define the wing windows as $200 < |v - v_{\text{sys}}| < 450$ km/s (blue and red separately) and measure the integrated flux and its uncertainty: $\sigma_{\text{int}} = \text{rms_channel} \times \Delta v_{\text{chan}} \times \sqrt{N_{\text{chan}}} \times \sqrt{(\text{aperture area} / \text{beam area})}$, with rms measured from the line-free channels of the same spectrum. The gate passes if the combined (blue+red, both lines considered) wing signal reaches 3σ in at least one of HCN(2-1) or HCO+(2-1).

Gate outcome. Pass $\rightarrow$ full three-transition ladder. Fail on either check $\rightarrow$ execute the pre-committed pivot: the science analysis uses only Bands 6 and 7 (3-2 and 4-3), and the Band 5 measurement enters the thesis as a quantified 3σ upper limit on the J=2-1 wing flux (if coverage passed but detection failed) or as explicitly stated missing coverage (if check A failed). Every subsequent step below is written for the full case; under the pivot, delete the 2-1 rows from each table and replace the two ladder ratios with the single 4-3/3-2 ratio, with conclusions framed as density-temperature bounds rather than tight constraints. No other change to the plan is permitted.

3. Common-resolution, common-grid cube set (weeks 2-3). Produce the master analysis cubes: for each of the six science lines (plus H¹³CN 4-3 and, as a bonus check only, CS 7-6), a subcube covering $v_{\text{sys}} \pm 600$ km/s, convolved to the common 7.5" beam, regridded to the common 10 km/s velocity grid and a common spatial grid (use the Band 5 pixel grid as reference), in Jy/beam. Convert aperture-integrated spectra to Jy (and, where useful for plots, to brightness temperature via the standard Rayleigh-Jeans conversion at each line’s frequency — never mix units silently). Measure and tabulate the per-channel rms of each master cube from line-free channels. From here on, all core/wing science numbers come from these master cubes and the single fixed aperture; nothing is measured from the native-resolution cubes except in steps 6 and 7.

4. Spectral decomposition (weeks 3-5). *Primary method — fixed velocity windows.* From the brightest, highest-S/N line (expected: HCN(4-3) or HCO+(3-2); choose once, by S/N, and record it), define three windows: core = $|v - v_{\text{sys}}| < v_{\text{c}}$, blue wing = -450 to -200 km/s, red wing = +200 to +450 km/s, with v_{c} set where the bright line’s profile falls to $\sim 10\%$ of peak or 200 km/s, whichever is smaller — the wing windows stay fixed at $\pm(200-450)$ km/s regardless. Apply these identical windows to every line of both molecules. For each line and window, record: integrated flux (Jy km/s), statistical uncertainty (formula from step 2C), and the number of channels 2σ within the window. Wings not detected at 3σ in a given line are reported as 3σ upper limits computed over the full wing window, and those limits propagate into the ratios as one-sided bounds.

Robustness check — multi-Gaussian fitting. Fit each aperture spectrum with up to three Gaussians (core + blue + red) using ‘lmfit’ or ‘astropy.modeling’, with the velocity centroids and widths of each component tied across all transitions of both molecules (i.e., fit the full set of spectra simultaneously with shared kinematic parameters, free amplitudes per line). Assess whether the wing components are statistically justified with a BIC comparison against the core-only fit. The

Gaussian wing fluxes are compared to the window-integrated fluxes; agreement within uncertainties is noted, disagreement beyond $\sim 1.5\sigma$ triggers a documented investigation of the baseline and window edges. The window-integrated values remain the headline numbers in all cases; the Gaussian results appear as a consistency column, never silently substituted.

5. Excitation and chemistry diagnostics (weeks 5–9). *Ratios.* For core, blue wing, and red wing separately (and blue+red combined for the weakest lines), compute in velocity-integrated flux units, then convert to the brightness-temperature ratios RADEX uses: HCN 3–2/2–1, 4–3/2–1 (and 4–3/3–2), the same for HCO+, and HCN/HCO+ at each matched J. Statistical errors propagate in quadrature; upper limits produce one-sided ratio limits.

Systematic floor. Attach to every cross-band ratio a flux-calibration systematic of 5% per Band 5/6 cube and 10% per Band 7 cube (standard ALMA Cycle 5 values — cite the ALMA Technical Handbook for the cycle), added in quadrature between the two bands entering each ratio, and further inflated by the resolved-out-flux floor from step 6. Ratios within a single band (e.g., HCN/HCO+ at matched J when both lines sit in the same cube) carry no cross-band calibration term — flag these in the tables as the calibration-robust subset, because they are.

RADEX grid. Pre-compute a non-LTE grid with RADEX (via the ‘spectralradex’ Python wrapper) for HCN and HCO+ separately: kinetic temperature 10–300 K (log-spaced, ~ 20 steps), $n(\text{H})$ 10^3 – 10 cm^{-3} (log-spaced, ~ 25 steps), and three fixed column-density-per-linewidth values spanning $N/\Delta v = 10^{13}$ – 10^1 $\text{cm}^{-2} (\text{km/s})^{-1}$, uniform sphere geometry, $T_{\text{bg}} = 2.73$ K, using LAMDA collision rates. For each (T_{k} , n) point, store the model 3–2/2–1 and 4–3/2–1 ratios. Interpretation is by overlaying the measured ratio bands (value $\pm$ total uncertainty) as contours on the (T_{k} , n) plane — one figure per component (core, blue, red) per molecule — and reading off the allowed region. Do not fit; bracket. The deliverable statement has the form “the wing ratios require $n(\text{H})$ in range X at any temperature in the grid, whereas the core requires range Y,” with the $N/\Delta v$ dependence shown by plotting all three column panels. Check the optically-thin/thick regime of each solution against the H^{13}CN constraint below.

Opacity check. Measure $\text{H}^{13}\text{CN}(4\text{--}3)$ in the core window from the Band 7 master cube; the HCN/ $\text{H}^{13}\text{CN}(4\text{--}3)$ ratio, compared against an assumed $^{12}\text{C}/^{13}\text{C}$ of ~ 40 – 70 (cite the adopted range), brackets (HCN 4–3) in the core. For the wings, compute the 3σ H^{13}CN limit and report the resulting lower limit on the ratio as an optical-depth constraint, labeled explicitly as a limit. Record how the core estimate does or does not move the RADEX-allowed region (RADEX handles the line opacity internally; the isotopologue check validates the chosen $N/\Delta v$ panels).

6. Flux-recovery systematic from internal Band 6 redundancy (weeks 9–10). Convolve the $\sim 0.8''$ Band 6 cube (‘Xb4d’) to the $\sim 4.4''$ Band 6 beam and regrid both to the common grid. Extract HCN(3–2) and HCO+(3–2) spectra through the identical fixed aperture from both, and compute the flux ratio ($0.8''$ -convolved / $4.4''$) in the core window and, if S/N permits, in the wing windows. The fractional flux deficit found (call it f) becomes the adopted floor on the resolved-out-flux systematic for all cross-band, cross-configuration ratios: add f in quadrature to the calibration systematic. Then rerun the RADEX overlay figures twice more — once with the systematic at the floor, once inflated to $2f$ — and record which conclusions survive each case. Any conclusion that dies at $2f$ is demoted in the write-up from a quantitative bracket to a sign statement (“wing gas more/less excited than core gas”) or dropped; this stress test is mandatory and its results are reported as a table of “conclusion vs. systematic level.”

7. Wing-origin check (hard-capped at two weeks, weeks 10–12). Using the native-resolution $\sim 4.4''$ Band 6 and $\sim 3.2''$ Band 7 cubes (not the master-smoothed set, and not the

0.24"/0.8" cubes), make integrated-intensity (moment-0) maps of the blue and red wing windows for the strongest wing-detected line, plus a moment-1 map of the core window to establish the nuclear rotation pattern and the kinematic major axis. Measure the centroid position of each wing moment-0 map by 2D Gaussian fit (or intensity-weighted centroid if the emission is not compact), with uncertainty from the standard $\sigma_{\text{centroid}} = \text{beam}/(2 \times \text{S/N})$ plus astrometric registration checked against the continuum peaks. The test: are the blue/red wing centroids displaced from the nuclei along a direction consistent with the minor axis (outflow-like), or do they lie along the major axis on opposite sides matching the rotation sense (rotation/tidal-like)? Compute the displacement, its significance, and its position angle relative to the kinematic axes. Whichever of the three outcomes occurs — outflow-like, rotation-like, or ambiguous — it is written up as-is, and the strength of the "wings = outflow" language in the interpretation is set accordingly. At the end of two calendar weeks, whatever state this check is in is the final state; no extension.

8. Write-up (weeks 12 onward, drafting begun in parallel from week 5). The thesis answers the single bounded question with a fixed set of products: (i) the gate-check documentation and outcome; (ii) one table of core/blue/red integrated fluxes and uncertainties for every line, including upper limits; (iii) one table of ratios with statistical and systematic errors separated, and the calibration-robust same-band subset flagged; (iv) the RADEX overlay figures per component per molecule, at three systematic levels; (v) the flux-recovery measurement f ; (vi) the wing-origin check result and the resulting caveat weight; (vii) a conclusions section that states the excitation contrast between wing and core gas at whichever confidence level survived the stress test, and what that contrast implies for the outflow's ability to form stars via the dense-gas conditions. Methods, scripts, and the notebook are included as reproducibility appendices. The seven analysis steps above are the complete scope; no analysis is added mid-stream.